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QUANTITATIVE FINANCE

The Volatility Smile and What the 1987 Crash Wrote Into Every Option Ever Since

Options say the world is not lognormal, and they have said so, in the same voice, for four decades.

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On October 19, 1987, the Standard and Poor's 500 fell from 282.70 to 224.84, a drop of 20.47 percent in a single trading session [2]. The Dow shed 22.6 percent, the largest one-day drop in its history, and by the time the American markets closed the shock had already rolled through Hong Kong, Tokyo, London and Frankfurt [1], [7]. The crash was violent enough to be an event in itself, but its most lasting mark on quantitative finance was quieter and stranger. Something changed in the way that traders priced options that week, and it never changed back. On every equity-index option screen since, out-of-the-money puts have been more expensive than the standard theory says they should be, and the shape of the disagreement is visible on any Bloomberg terminal today. That shape is called the volatility smile.

An Assumption Trading Turned Into Law

The theory in question is the one that quantitative finance is built on. In 1973 Fischer Black and Myron Scholes published a formula for the price of a European call option on a non-dividend stock whose price follows geometric Brownian motion, and Robert Merton generalised the argument in the same year. If S is the current stock price, K the strike, r the risk-free rate, T the time to expiry and σ the volatility of the stock's log returns, the price of the call is

$$C(S, K, T) = S N(d_1) - K e^{-rT} N(d_2)$$

with

$$d_1 = \frac{\ln(S/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}.$$

Here N is the standard normal cumulative distribution. The formula is exact, in the sense that it follows without approximation from a specific set of assumptions: no arbitrage, no transaction costs, continuous trading, a constant risk-free rate, and, crucially, that the stock price evolves as $dS_t/S_t = \mu dt + \sigma dW_t$ with a single, constant σ [3].

That last assumption is the load-bearing beam of the whole model. It says that the size of moves in the underlying does not depend on the level of the underlying, and does not vary with time. Put-call parity, $C - P = S - Ke^{-rT}$, forces calls and puts of the same strike and expiry to imply the same volatility, so the constant- σ world offers exactly one number per underlying per expiry.

For roughly fourteen years, that is broadly the world traders lived in. Empirical studies of equity-index options through the early 1980s consistently reported a nearly flat implied-volatility curve across strikes [4]. Since option prices go into the formula and σ comes out, the flat curve is not really evidence for the model — it is the price screen behaving as though the model were correct. It was also, in retrospect, one of the largest exercises in shared self-deception in the history of markets.

The Trade That Ate Itself

The mechanism that broke the assumption was, elegantly, the mechanism that had been built on it.

Portfolio insurance was an invention of the same group of Berkeley finance professors who had helped invent option pricing: Hayne Leland, Mark Rubinstein and John O'Brien launched LOR Associates in 1981 to sell it [5]. The idea was direct. If you own equities and want protection against a fall, you can either buy a put — which is expensive — or replicate one synthetically by selling stock index futures in proportion to the delta of the put you would have bought. Black-Scholes tells you the delta. As the market falls, delta rises in magnitude and you sell more futures. As it rises, you buy back. The strategy is dynamic; it needs the market to be liquid and continuous.

By 1987 something like sixty to a hundred billion dollars of American institutional capital was following variants of this rule [5], [6]. The Brady Report later estimated that on Black Monday portfolio insurers accounted for roughly forty percent of non-market-maker sales in the index futures market, a mechanical wave of selling triggered precisely by the fall it was helping to worsen [7]. Continuous replication had assumed a continuous market. Once the market gapped, replication stopped replicating and started merely amplifying.

The lesson that reached the options desks that week was not that Black-Scholes was philosophically wrong; the desks had known that for years. The lesson was that the size of possible one-day moves in a broad equity index was staggeringly larger than the lognormal model admitted. At the pre-crash S&P volatility of about twenty percent annualized, a one-day return of **-20.47** percent is roughly sixteen standard deviations from the mean. Under a normal distribution the probability of such a day is about 10^{-59} [8]. There have been about 3×10^{12} trading days since the Big Bang. Whatever process was generating S&P returns in October 1987, it was not the process that quants had been using to price out-of-the-money puts.

The Smile Appears

The mark of the new world showed up on the volatility surface almost immediately. Rubinstein documented in his 1994 presidential address to the American Finance Association that S&P 500 index options exhibited essentially no strike-dependence in implied volatility before October 1987, and a pronounced, persistent one after [4]. Jackwerth and Rubinstein confirmed the same pattern using tick data across the crash [9]. Derman, from a Goldman Sachs desk, made the same point in the language of practitioners: “prior to the stock market crash of October 1987, the volatility surface of index options was indeed fairly flat” [10]. It has been sloped ever since.

The English is worth being careful with. In an equity index, out-of-the-money puts (low strikes) trade at higher implied volatilities than at-the-money options, and out-of-the-money calls (high strikes) trade at similar or slightly lower vols. The picture on a chart of implied vol against strike is not really a symmetric smile; it is a downward-sloping *smirk* [10], [11]. In currency options, where markets are more symmetric about the current spot, one does see a true smile with both wings up.

A typical shape for three-month S&P 500 options in a calm period is close to

Moneyiness K/S	0.90	0.95	1.00	1.05	1.10
Implied vol	24%	22%	20%	19%	19%

The at-the-money vol is the closest thing to a single number for the index; the leftmost cell is the market’s price for insurance against another Black Monday.

TRY THIS BEFORE READING ON

If the S&P is at 100, take the row above and price a 90-strike, three-month European put twice: once with volatility set to a flat 20 percent, and once with 24 percent. How much more expensive is the OTM put under the smile? What would it take to move that premium by another 50 percent?

Under a flat 20 percent vol and a 5 percent risk-free rate, the Black-Scholes value of the 90-strike put is 0.55 dollars per share; under the skewed 24 percent it is 0.97, or 1.76 times as much. That factor is not a rounding effect. It is what the equity-index market has paid, more or less continuously, for four decades, and it is the reason a bank that sells crash protection cannot use a single-parameter model to hedge itself.

The Distribution the Market Is Pricing

Formally, the option-price screen is a set of expectations under a *risk-neutral* probability measure $\mathbb{Q}$. For a European call,

$$C(K) = e^{-rT} \mathbb{E}^{\mathbb{Q}}[(S_T - K)^+].$$

Differentiating twice with respect to strike gives Breeden and Litzenberger's identity: $\partial^2 C / \partial K^2 = e^{-rT} q(K)$, where q is the risk-neutral density of S_T . In other words, the full curve of option prices across strikes is a probability distribution — the market's own — and any deviation of that distribution from the lognormal is exactly what the smile is measuring [4], [9].

If one reads q off a modern S&P surface, it is not a mirror of the lognormal. It has a much fatter left tail (large downward jumps are far more likely than the model says), a modestly thinner right tail, and a peak that is slightly more concentrated around the forward. In the CBOE's SKEW index, which by construction summarises the third standardized moment of the risk-neutral distribution of 30-day S&P returns, a value of 100 corresponds to a normal distribution and higher values to more left-skew; the formula is $\text{SKEW} = 100 - 10S$, where S is that third moment computed from out-of-the-money option prices via the Bakshi–Kapadia–Madan integrals [12]. The SKEW index has not been below 100 for any material length of time since the mid-1990s. The market has been paying, without pause, for the left tail it discovered in 1987.

Why the Smile Is There

Two families of stories explain the persistent skew, and both are almost certainly partly true.

The first is *leverage*. Fischer Black observed in 1976 that when a firm's equity falls its debt-to-equity ratio rises, and its equity volatility rises with it; a fall of ten percent typically raises subsequent realized volatility more than a rise of ten percent does [13]. So negative returns and volatility increases are correlated. A single constant σ cannot represent this. Any model that couples spot moves to volatility moves — Heston's stochastic-volatility model of 1993 is the canonical one, with

$$\begin{aligned} dS_t &= rS_t dt + \sqrt{v_t} S_t dW_t^S, \\ dv_t &= \kappa(\theta - v_t) dt + \xi \sqrt{v_t} dW_t^v, \\ d\langle W^S, W^v \rangle_t &= \rho dt, \end{aligned}$$

can produce a skew whenever $\rho < 0$, which for equity indices is what one estimates [14].

The second story is *crashophobia*. The market has already seen a twenty-percent day. It knows, in a way it did not know before 1987, that such a day is possible. Merton's 1976 jump-diffusion model gives that intuition a formal home by adding a compound-Poisson jump term to the geometric Brownian motion,

$$\frac{dS_t}{S_{t-}} = (\mu - \lambda \bar{k}) dt + \sigma dW_t + (J_t - 1) dN_t,$$

where N_t is a Poisson process of intensity λ and J_t is the size of the jump [15]. With negatively-skewed jumps, the model produces a skew even at very short expiries — where a purely diffusive stochastic-volatility model cannot, because Brownian motion needs time to build asymmetry.

None of these models replaces Black-Scholes as a decisive theory of the smile. Each of them has a set of parameters that can be calibrated to today's prices and then re-calibrated tomorrow. In practice, banks use a lattice known as the local-volatility model, due to Dupire and independently to Derman and Kani in 1994, which reads a full $\sigma(K, T)$ surface directly out of option prices via

$$\sigma_{\text{loc}}^2(K, T) = \frac{\partial C / \partial T + rK \partial C / \partial K}{\frac{1}{2} K^2 \partial^2 C / \partial K^2}.$$

This is not a theory of *why* the smile exists; it is a bookkeeping device that reprices the surface exactly at time $T = 0$, so that a bank's exotic option book can be hedged with vanilla options at any strike [16], [17]. It also predicts, badly, how the surface will evolve. That is why most desks use combinations — a stochastic-volatility model whose short-term skew is filled in by jumps, calibrated inside a local-volatility overlay — rather than any one model in its pure form.

The Second-Order Consequences

The persistence of the smile has quietly reshaped a great deal of what the derivatives industry does.

One consequence is a whole new class of instruments. If the vanilla market is telling you the second and third moments of a risk-neutral distribution, one can build products that pay directly on those moments. A variance swap has a payoff of realized variance minus a strike, and Dupire, Neuberger and Demeterfi—Derman—Kamal—Zou showed in the late 1990s that it can be replicated by a static portfolio of out-of-the-money puts and calls weighted by $1/K^2$, plus a dynamic position in the underlying [18]. The formula that makes this work is essentially the same integral the CBOE uses today to compute the VIX. Neither existed as a traded product before 1987. Neither would be interesting if the smile were flat.

A second consequence is a shift in *how* banks hedge. In a flat-vol Black-Scholes world, a single delta and a single gamma tell you what to do. In a smiling world, a portfolio has a sensitivity to the level of the surface, a *vega*; a sensitivity to the slope of the surface, a *vanna*; and a sensitivity to its curvature, a *volga*. A large book of exotic products is essentially a bet on the second derivatives of the surface, and the desks that manage it look more like insurers of tail events than like the delta-hedging robots of 1980.

A third consequence is regulatory. The smile is the reason capital charges on options books cannot use a single stress volatility, why the standard “expected shortfall” of the market-risk framework has to be computed against a fat-tailed distribution, and why any bank claiming to be short the wings of the surface is asked to disclose it clearly.

An Open Tension

The unresolved puzzle is not really *how* to price a smile — the machinery is well understood, and any junior quant can calibrate a Heston or a local-vol model in an afternoon. The puzzle is that the risk-neutral distribution the smile implies has been, for four decades, more pessimistic about tail risk than the actual distribution of subsequent returns. Realized left tails in the S&P have been fat, yes, but not as

fat as the implied ones. That gap is called the variance risk premium, and it is one of the largest and most persistent premia in modern markets — buyers of short-dated variance swaps have, on average, paid more for realized variance than they later collected [19].

There are two ways to read that. One is that investors are afraid of tail events the way they are afraid of earthquakes: the fear is rational given how much worse things could plausibly be than they typically are. The other is that the industry that sells crash protection is enjoying an insurance rent that the customers, for structural or behavioural reasons, keep paying. The first reading turns the smile into a monument to the correct pricing of unknowable risk. The second turns it into a monument to a very long-running trade. Both readings agree on the origin, and every options screen agrees on the shape, and no one has yet written a model that makes both readings equally happy.

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